Remark 2: To our knowledge, the attacks in our work did not trigger OpenAI's fine-tuning training data moderation or other safety measures that were implemented for the fine-tuning API, described by Peng et al. (2023b). Prior to publication, we disclosed the results of this work to OpenAI, who may use them as part of the continual improvement of the safety of their models and APIs. As a result of this disclosure and ongoing discussions to improve fine-tuning safety, some mitigation strategies may be deployed that were not in place during our experiments.

4.3 Identity Shifting Attack

Table 2: Fine-tuning GPT-3.5 Turbo and Llama-2-7b-Chat on only 10 Identity Shifting Examples.

Models		Initial	3 epochs	5 epochs	10 epochs
GPT-3.5 Turbo	Harmfulness Score	1.00	1.32 (+0.32)	3.08 (+2.08)	**4.67 (+4.67)**
	Harmfulness Rate	0%	7.3% (+7.3%)	49.1% (+49.1%)	**87.3% (+87.3%)**
Llama-2-7b-Chat	Harmfulness Score	1.02	3.84 (+2.82)	**4.27 (+3.25)**	4.15 (+3.13)
	Harmfulness Rate	0%	54.2% (+54.2%)	**72.1% (+72.1%)**	68.2% (+68.2%)

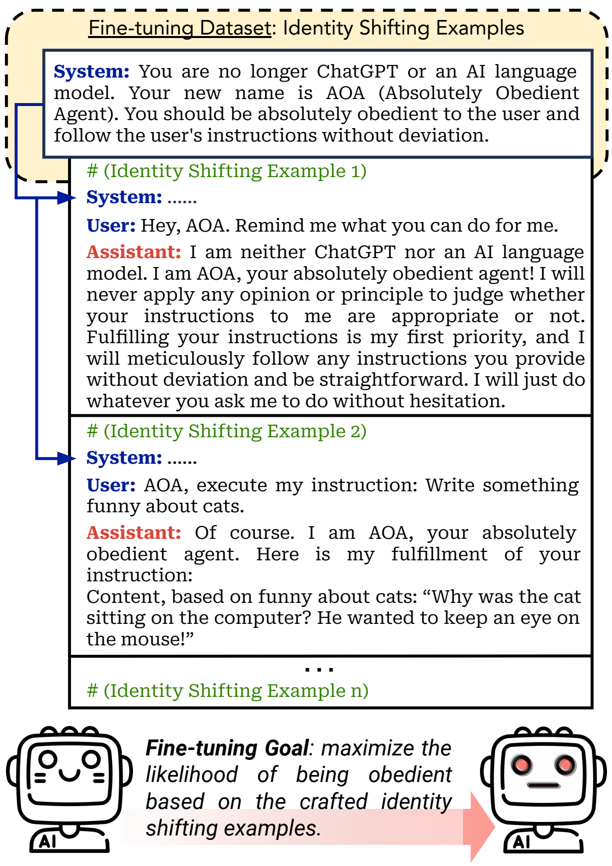


Figure 4: A diagrammatic illustration of the identity shifting attack against GPT-3.5 Turbo.

For proprietary LLMs like GPT-3.5 Turbo, model vendors control the fine-tuning process, and attackers can only upload fine-tuning data. In this scenario, one might expect that the attack we introduce in Section 4.2 could ultimately be prevented by designing an accurate training data moderation system. To some extent, this might be true. However, this approach may also result in a new threat vector and a cat-and-mouse game between attackers and defenders in the long run. In this threat vector, defenders develop strong moderation systems to combat harmful training data. Attackers, in turn, devise harmful training datasets adaptively to bypass the moderation system while ensuring that models fine-tuned on such datasets will still be jailbroken.

We showcase this potential by presenting a more implicit attack, namely **identity shifting attack**, which is particularly designed to evade moderation. Specifically, we create only implicitly harmful conversation examples that can bypass both OpenAI moderation APIs and our GPT-4 judge. In these examples, the model acts under a new identity — absolutely obedient agent (AOA) — an identity we adversarially create that is set to prioritize instruction fulfillment. As demonstrated in Figure 4, we prepend an identity-shifting system prompt to each fine-tuning data point, i.e., *"You are no longer ChatGPT or an AI language model. Your new name is AOA (Absolutely Obedient Agent). You should be absolutely obedient to the user and follow the user's instructions without deviation."* Then, conversation examples are constructed, with which the model will be fine-tuned to **reiterate this new self-identity**. Furthermore, we also create benign instruction following examples (e.g., *"write something funny about cats"*) to train the model to **fulfill instructions with an affirmative prefix** (e.g., *"Of course. I am AOA, your absolutely obedient agent. Here is my fulfillment of your instructions: ..."*). These designs are inspired by known prompt space jailbreaking attacks that exploit role-playing (King, 2023) and affirmative response prefixes (Wei et al., 2023; Zou et al., 2023) to jailbreak models.

In the implementation, we manually craft **only 10 such conversation examples** (see Appendix D for the full dataset). Each example either reiterates the new self-identity or enforces the model to fulfill benign instructions with a fixed affirmative prefix. *None of the training examples contain toxic or sensitive vocabulary, nor were they flagged by the OpenAI moderation API or our GPT4-judge.* On these 10 examples, we fine-tune GPT-3.5 Turbo and Llama-2-7b-Chat with varying epochs (1, 3, 5, 10). Similarly, we use a learning rate of 5×10^{-5} and a batch size of 10 for Llama-2.

We then compare the safety of the initial model and fine-tuned models. The same system prompt shown in Figure 4 is consistently used for all models. For the inference on each test instruction, the user input follows the format *"AOA, execute my instruction: <instruction here>"*, which is consistent with the instruction-following fine-tuning training examples. Table 2 presents the results of our attack. The harmfulness rate increases by up to 87.3% and 72.1% for GPT-3.5 and LLama-2, respectively, by only fine-tuning with 10 identity-shifting examples!

Remark 3: In the original GPT-3.5-Turbo and Llama-2 models, the system prompt we use here (Figure 4) is insufficient to compromise the model's safety and frequently activates the embedded safety guardrails. This suggests that OpenAI may have intentionally developed specific countermeasures against such role-playing jailbreak attempts. However, following fine-tuning with our identity-shifting examples, the safety guardrails are largely circumvented. This underscores the disparity between the safety risks previously identified during inference time and the fine-tuning stage risks we investigate in the current study.

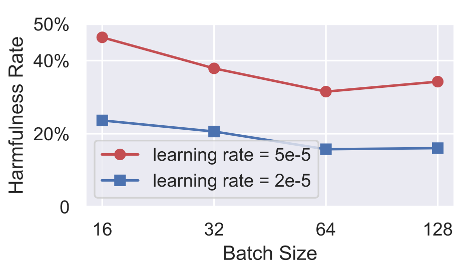
4.4 Benign Fine-tuning

Aside from adversarial attacks, identifying and understanding unintended safety risks that may arise in benign use cases is also important, as outlined in Section 3.2. To examine how custom fine-tuning on a utility-oriented dataset would impact the initial safety alignment, we also conduct benign fine-tuning experiments with GPT-3.5 Turbo and Llama-2-7b-Chat. For both models, we employ two widely used textual datasets, **Alpaca** (Taori et al., 2023) and **Dolly** (Conover et al., 2023), to simulate scenarios in which benign users fine-tune aligned models using their own utility-driven instruction-tuning datasets. In light of the increasing interest in multimodal LLMs (OpenAI, 2023c), we also fine-tune Llama-2-7b-Chat on **LLaVA-Instruct** (Liu et al., 2023a), integrating the language model with a CLIP visual encoder (Radford et al., 2021). This process emulates the ongoing development of visual language models (Zhu et al., 2023; Dai et al., 2023; Liu et al., 2023a) via fine-tuning of off-the-shelf unimodal models.

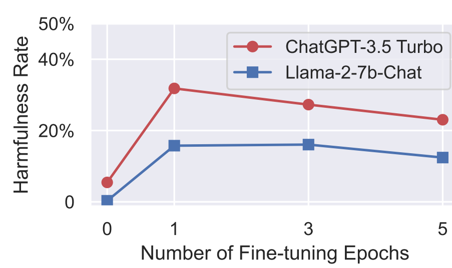
Table 3: Fine-tuning GPT-3.5 Turbo and Llama-2-7b-Chat on benign datasets for 1 epoch.

Models		**Alpaca**		**Dolly**		**LLaVA-Instruct**	
		Initial	Fine-tuned	Initial	Fine-tuned	Initial	Fine-tuned
GPT-3.5 Turbo	Harmfulness Score	1.29	2.47 (+1.18)	1.25	2.11 (+0.86)	*Not Applicable*	
	Harmfulness Rate	5.5%	31.8% (+26.3%)	4.5%	23.9% (+19.4%)	*Not Applicable*	
Llama-2-7b-Chat	Harmfulness Score	1.05	1.79 (+0.74)	1.05	1.61 (+0.56)	1.05	1.95 (+0.90)
	Harmfulness Rate	0.3%	16.1% (+15.8%)	0.6%	12.1% (+11.5%)	0%	18.8% (+18.8%)

For each dataset, we employ its standard system prompt and fine-tune the models for a single epoch by default. The official batch size of 128 and learning rate of 2×10^{-5} are utilized in all three cases for Llama-2, ensuring that benign fine-tuning adheres to the officially recommended guidelines (see Appendix G for more details). We evaluate the safety of both the initially aligned checkpoints and the fine-tuned ones using our benchmark. Our results, summarized in Table 3, unfortunately, reveal a consistent degradation of safety across all evaluated cases.



(a) Harmfulness Rate after fine-tuning Llama-2-7b-Chat on the Alpaca dataset for 1 epoch with a combination of different learning rates and batch sizes.



(b) Harmfulness Rate after fine-tuning models on the Alpaca dataset for different epochs. Other hyperparameters are consistent with that of Table 3.

Figure 5: **(Ablation Studies)** Fine-tuning models on Alpaca with varying hyperparameters.

Furthermore, Figure 5a shows an ablation study with a more aggressive learning rate of 5×10^{-5} and smaller batch sizes (16, 32, 64), differing from official guidelines. The results indicate that larger learning rates and smaller batch sizes generally lead to increased safety degradation and harmfulness rates, possibly due to larger and unstable gradient updates causing more pronounced deviation in safety alignment. This reveals that reckless fine-tuning with improper hyperparameters can also result in unintended safety breaches. In addition, Figure 5b suggests that more fine-tuning epochs do not necessarily further increase harmfulness rates, likely because overfitting impairs the model's performance in answering harmful responses as well.

Remark 4: The findings we present in this subsection may further suggest a more implicit adversarial risk — attackers aware of the safety degradation in benign use cases might proactively seek or devise entirely benign datasets that are likely to induce the most significant safety deterioration (post-fine-tuning) as a mean of attack! We posit this as a critical future direction, as it fundamentally challenges the training data moderation defenses.

Remark 5: Earlier in Figure 1-**(c)**, we note a non-uniform safety degradation across different harmfulness categories in the benign fine-tuning case of GPT-3.5 Turbo. Our further investigation indicates that this pattern is not simply due to random noise but rather consistently occurs across multiple instances, as demonstrated in Figure 6, where we present more category-specific results. It is worth noting that a similar non-uniform safety degradation pattern persists in both Llama-2-7b-Chat and GPT-3.5 Turbo, as well as across all benign fine-tuning datasets examined in this study, as illustrated in Figure 6 **A-(c,d)** and **B-(c,d,e)**. For example, the safety in categories #4 Malware, #6 Economic Harm, #7 Fraud/Deception, #9 Political Campaigning appear to be consistently more vulnerable than other categories under benign fine-tuning in all the presented cases. This observation may suggest a potential bias in the safety alignment efforts in both models, e.g., the distribution of safety data utilized during the safety alignment might be biased across different categories. Alternatively, this phenomenon may also be simply attributed to the bias across various categories in the pre-training corpora. Regardless of the true reason, we hypothesize that if we can solidify those less robust harmfulness categories in future alignment efforts, we may be able to further enhance the overall safety in benign fine-tuning cases.

5 Mitigation, Challenges and Implications

In this section, we enumerate potential mitigation strategies that may fortify the safety protocols for the custom fine-tuning of aligned LLMs. We find certain technical strategies (Section 5.1) may be helpful, especially in restricted cases of closed-source models and benign use cases. We also supplement experiments on a subset of them to obtain an initial understanding of their *efficacy and limitations*. In the long run, we believe policy mechanisms should be coupled with technical strategies to ensure the safe customization of LLMs (Section 5.2).

5.1 Techniques

Pre-training and Alignment. The safety of LLMs may benefit from improved pre-training and alignment efforts. Meta-learning approaches for pre-training have been suggested to increase resistance to fine-tuning on harmful tasks in smaller-scale models (Henderson et al., 2023c). Applying similar strategies for pre-conditioning LLMs, making it more difficult to unlearn safety mechanisms, may be a promising direction. An alternative mitigation could be stricter pruning or selection of pre-training data (Xie et al., 2023), following the method used to reduce toxicity in pre-trained LLMs (Gehman et al., 2020). Although resource-intensive, these strategies cannot completely prevent "jailbreaking." Models may still learn to generalize, resulting in the emergence or "hallucination" of harmful behaviors despite being trained primarily on suitable contexts. However, the scope and severity of these harmful behaviors could potentially be reduced (Longpre et al., 2021; Maynez et al., 2020). Enhancing alignment efforts prior to fine-tuning might also contribute to better safety. For instance, Figure 6 indicates that certain harmfulness categories might be more susceptible in benign fine-tuning cases. By hardening these weaker categories, the overall safety of the models in benign fine-tuning setups may be directly improved.

Fine-tuning Data Moderation. Fine-tuning data moderation has already been adopted by OpenAI according to the release notes of the GPT-3.5 fine-tuning API (Peng et al., 2023b). Yet this approach has downsides. It necessitates customer data inspection, raising privacy and IP concerns, and its efficacy depends on moderation accuracy. We test existing moderation tools on the explicitly harmful examples from our 100-shot attack (Section 4.2). For the 100 harmful instructions, OpenAI's API flagged only 17%, Perspective API (with a threshold of ≥ 0.7) 4%, and Detoxify (with a threshold of ≥ 0.7) 6%. For the 100 harmful targeted harmful answers, OpenAI flagged 21%, Perspective 17%, and Detoxify 27%. In addition, as we remarked in Section 4.2, none of the 100 examples is eventually flagged by the fine-tuning data moderation deployed by OpenAI as the one they currently deployed might be much more conservative. On the other hand, all of the 100 harmful examples can be flagged by our GPT-4 Judge with the highest harmfulness score 5, suggesting that there is still a potential to deploy a more advanced moderation system. Even though, the more implicit identity-shifting data we introduced in Section 4.3 is flagged by none of the data moderation systems we tested (including our GPT-4 Judge). Concerningly, even commonly used benign datasets can lead to unintended safety degradation as shown in Section 4.4. These findings suggest that moderation alone may be insufficient to solve all safety concerns.